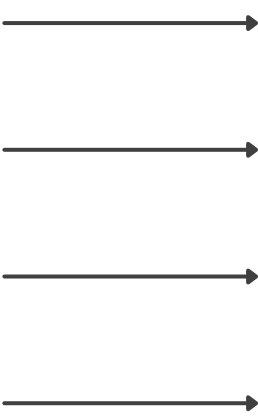


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<u>Problem:</u> Subjective manual grading at mandis causes an estimated 15-20% value loss for farmers, and inspection delays add to 25-30% post-harvest loss. Farmers accept whatever price is offered because they cannot prove quality to distant buyers.	<u>Our Idea:</u> KisanSetu AI is a mobile computer-vision platform that grades crops instantly from a phone photo, sets fair ML-based prices and links farmers directly to buyers through spot and future-harvest contracts.
<u>Proposed Solution:</u> • AI Produce Scanner: instant Grade A/B/C via smartphone camera • Dual Marketplace: Spot Market and Future Harvest forward contracts • Dynamic Pricing Engine: transparent price forecasts linked with e-NAM	<u>Innovation / Uniqueness</u> • Edge-based offline scanning: for low-connectivity rural zones • Immutable digital quality certificates that any buyer can verify • Grading, pricing and market access combined in one farmer-first flow

CURRENT MANDI FLAWS
✗ Subjective manual grading
✗ Inspection delays and spoilage
✗ Opaque, middleman-driven prices
✗ No proof of quality for buyers



OUR AI SOLUTION (KISANSETU AI)
✓ Objective AI grading in seconds
✓ Instant on-device scan, works offline
✓ ML price forecast synced with e-NAM
✓ Tamper-proof digital certificate

Software

User Authentication: Secure registration and profile verification for all users.

Produce Listing: CheckPoint-1 manages inventory and the essential details.

Marketplace Access: Buyers browse, compare, and place orders.

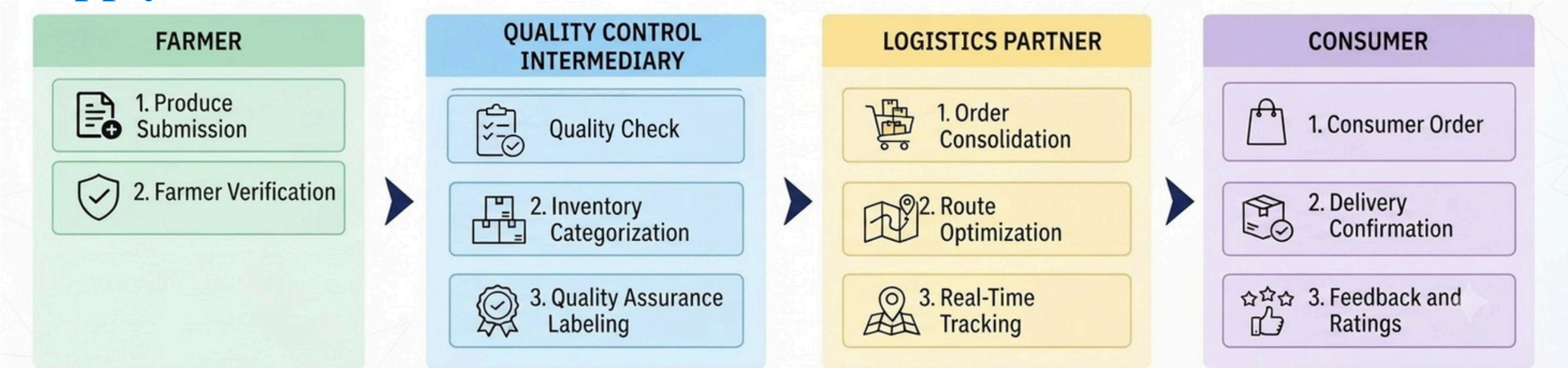
Order Management: Orders are validated and processed efficiently.

Payment & Fulfillment: Secure payments and coordinated delivery.

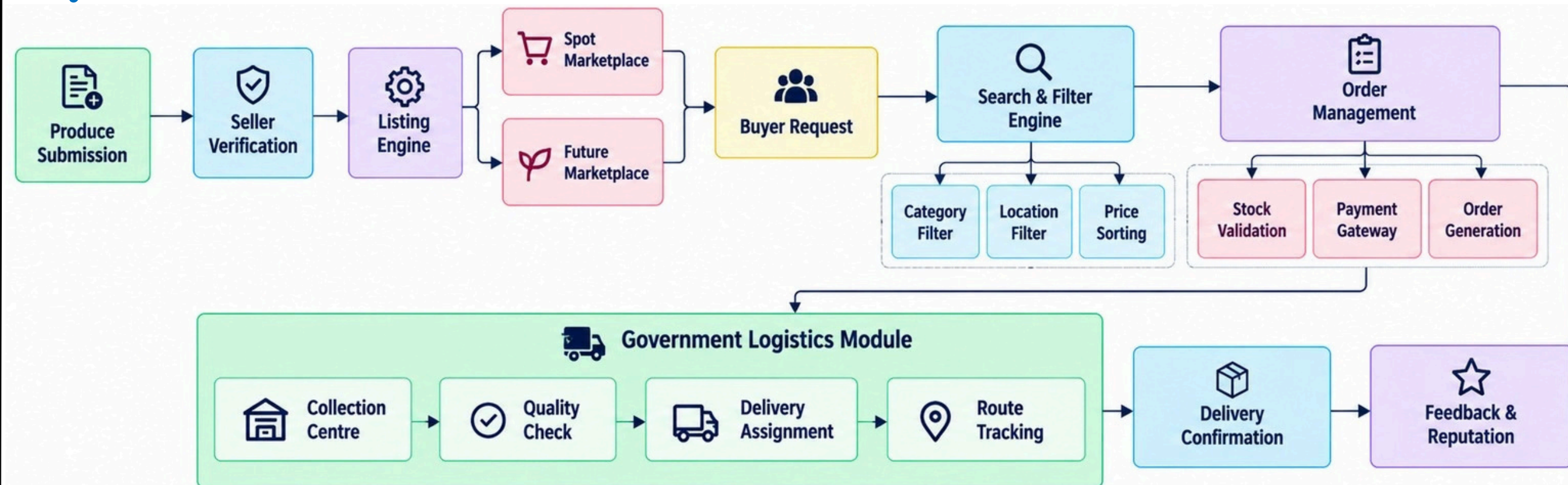
Order Tracking: Real-time tracking until successful delivery.

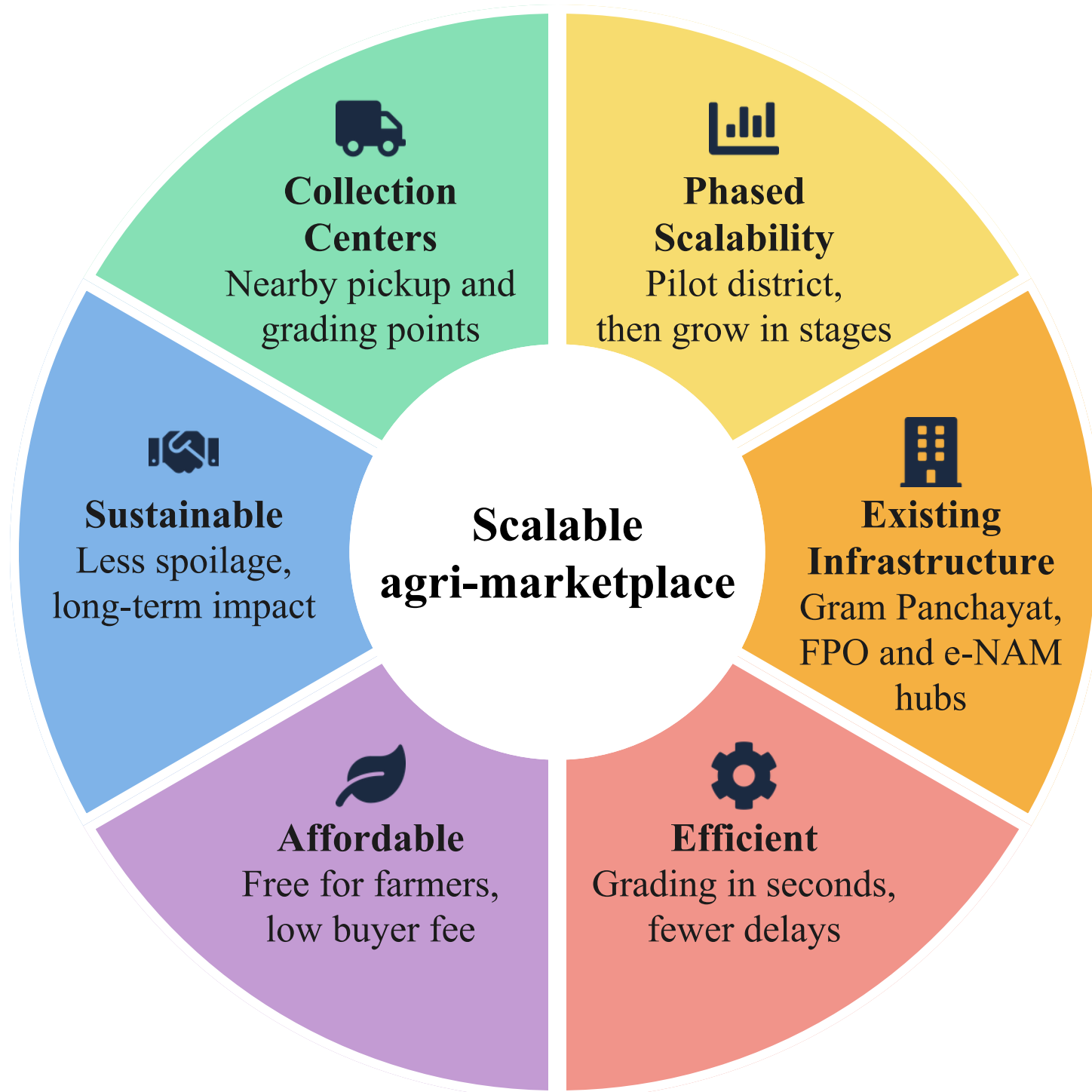
Feedback System: Ratings and reviews improve platform trust.

Supply-Chain Flow:



System WorkFlow:





Feasibility Matrix

Technical

Lightweight on-device AI model (<25 MB) runs on entry-level Android phones

Operational

Uses existing Gram Panchayat and FPO collection hubs; no new buildings

Financial

0% fee for farmers; 1-2% fee only for institutional buyers

Challenges & Strategies

Lighting & background variation
Photos vary in light and backdrop

Image Normalization

Auto-normalise images and show boundary overlays in the camera

Low smartphone literacy
Many farmers are new to apps

Voice-Guided Access

Multilingual voice-guided UI plus IVRS for basic phones

Buyer trust
Buyers doubt remote quality claims

FPO-Backed Disputes

Digital certificates and an FPO dispute-resolution workflow

Government reforms and schemes

e-NAM Integration- Builds upon India's push toward transparent, unified digital agricultural markets.

APMC Reforms- Supports e-trading, wider markets and competitive price discovery.

Stronger Rural Economy- Increased farmer earnings boost rural purchasing power.

₹3.19L Cr → ₹4.84L Cr in cumulative e-NAM trade- rapidly growing role of Indian agri-Tech markets.

For Scalability

Multi-Stakeholder Marketplace:

Scales across farmers, consumers, retailers, restaurants and institutional buyers.

Regional Adaptability: Multilingual and region-specific features enable utilization across diverse agricultural communities.

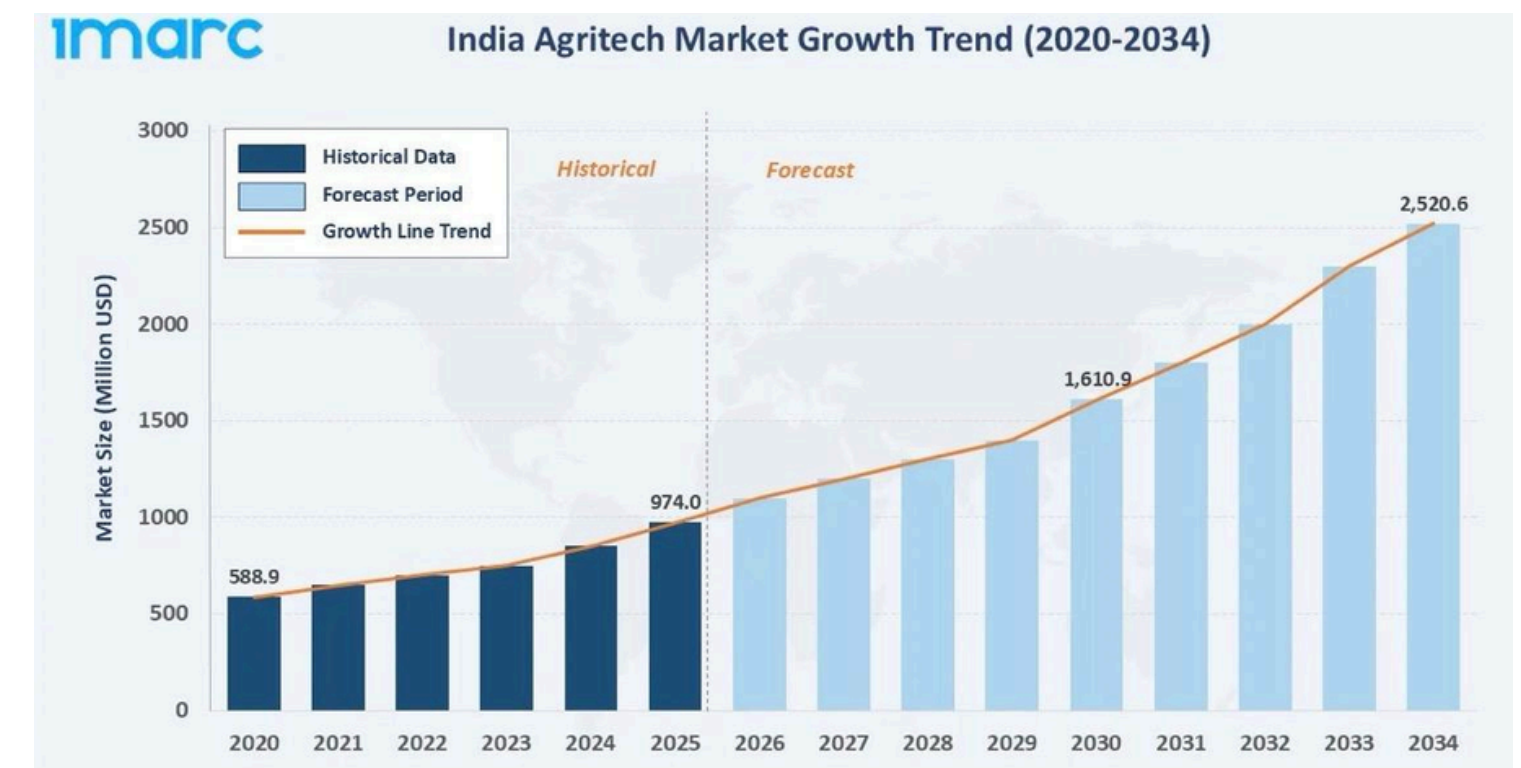
For farmers and consumers

Higher Earnings- Better price discovery, fewer unnecessary margins.

Fair Pricing- Reduced supply-chain inefficiencies and excessive markups.

Availability & Transparency- Efficient logistics and traceable transactions build access and trust.

Automated Quality Assessment- AI/ML enables consistent produce quality evaluation.



Links

● **e-NAM - Government of India**

● **National Agriculture Market - e-NAM**

● **Press Information Bureau e-NAM 2026**

● **NITI Aayog - Agri-Entrepreneurs & Innovators**

● **Agmarknet - Agricultural Market Information**

● **ONDC - Open Network for Digital Commerce**

Research Papers

● **RBI - Supply Chain Dynamics and Food Inflation in India**

● **Das et al. - Transforming Agricultural Trading and Commerce in India**

● **Swain et al. - Impact of Development of (e-NAM) on Farmers**

● **Richa Kumar - Elusive Empowerment: Information Provision & Disintermediation in Soybean Markets**